Video 3

LangChain Vs LangGraph

Goal

1. Why LangGraph exists?
2. What is LangGraph?
3. LangChain vs LangGraph

LangChain

→ LangChain is an open ~~based~~ source library designed to simplify the process of building LLM based applications.

↳ It provides modular building blocks that let you create sophisticated LLM-based workflows.

CamScanner

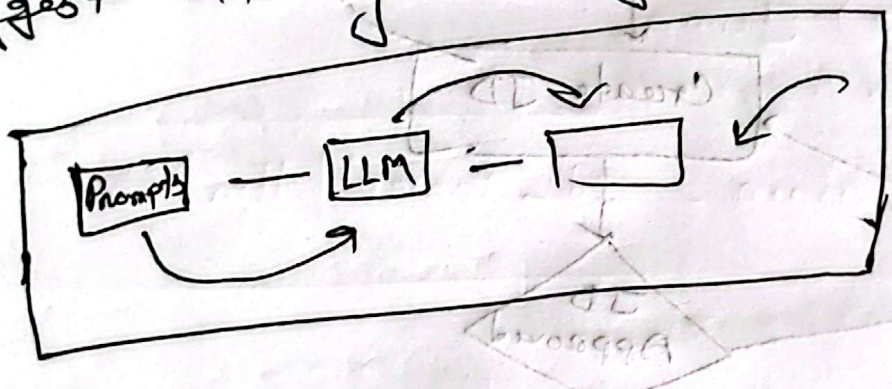
Components

Model

Prompts

Retrievers

the biggest offering of Langchain is chains.



What can you do with Langchain

→ What can you do with Langchain

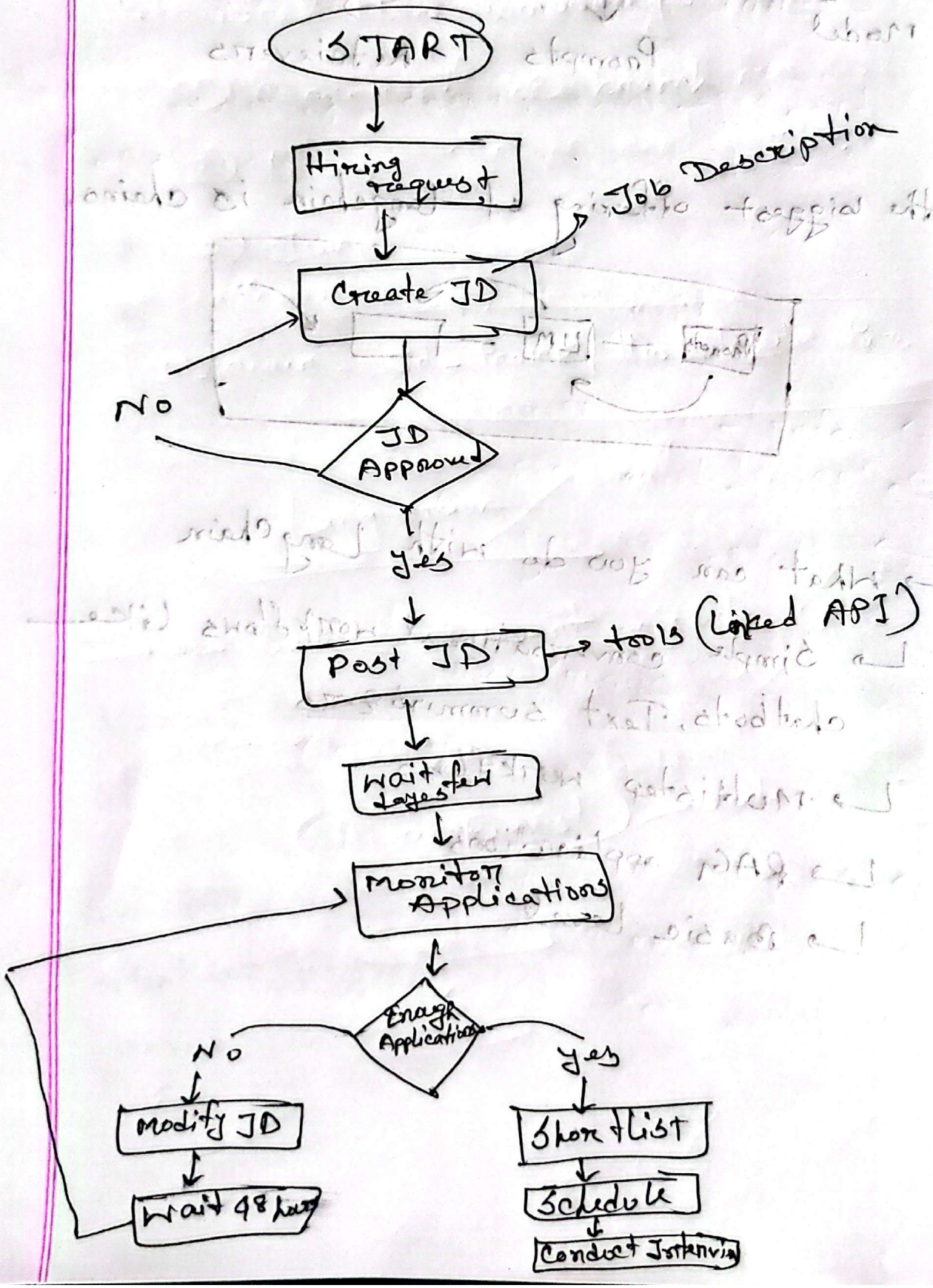
↳ Simple conversational workflows like chatbots, Text summarizers

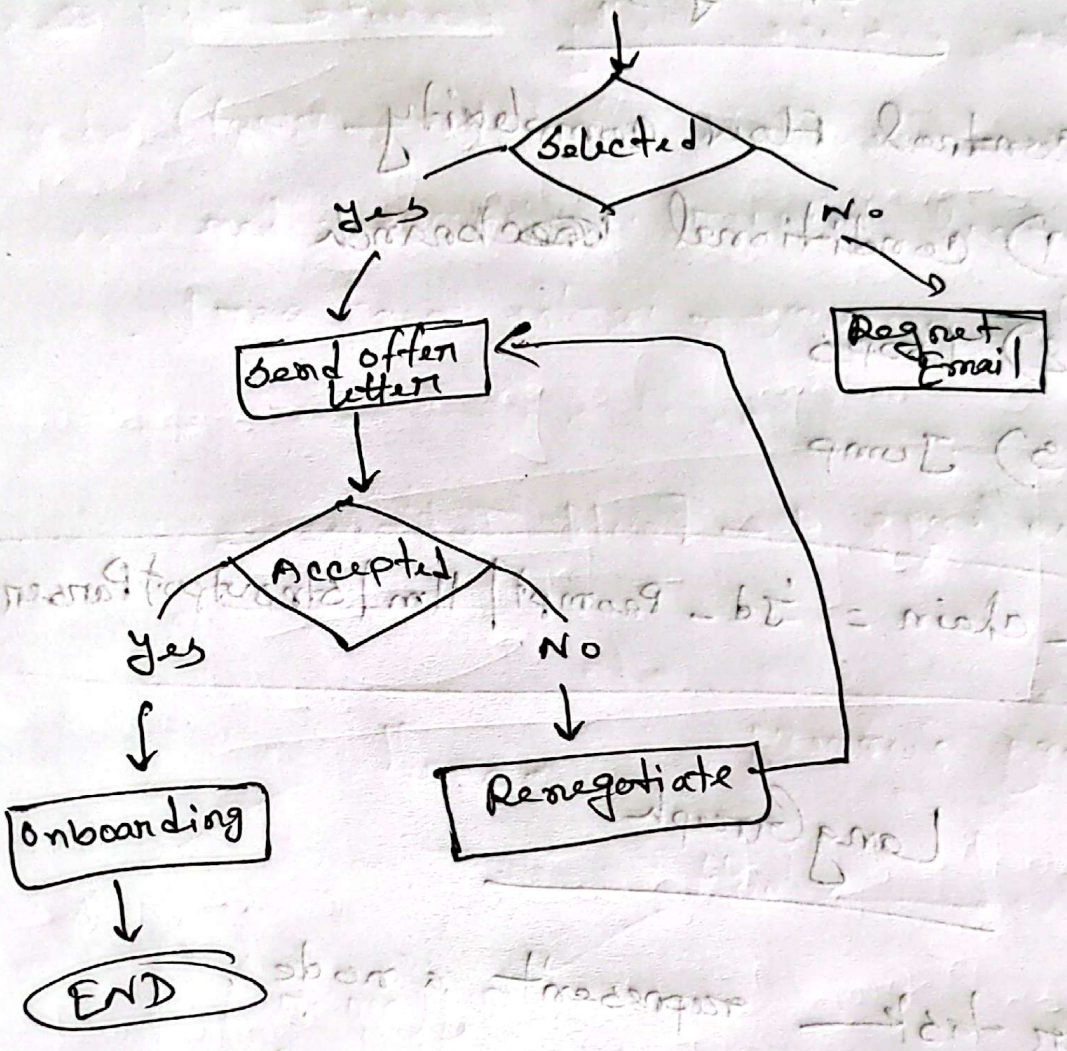
↳ Multistep workflows

↳ RAG applications

↳ Basic level

Automated Hiring Workflow





→ In this full workflows implement a code that is LangChain.

chalenges

1. Control Flow complexity
   1) conditional branch
   2) Loops
   3) Jump

```
jd_chain = jd_Prompt | llm | strOutputParser()
```

LangGraph

→ Per task represent a node.

supposes that,

HiringRequest → node.

↓

Jd

↓

→ challenge 2 – Handling State

LangGraph

↓

State object → Pydantic

State object → Typedictionary

LangChain → stateless

LangGraph → stateful

→ challenge 3 – Event Driven Execution

Workflow

Sequential

event driven

↓

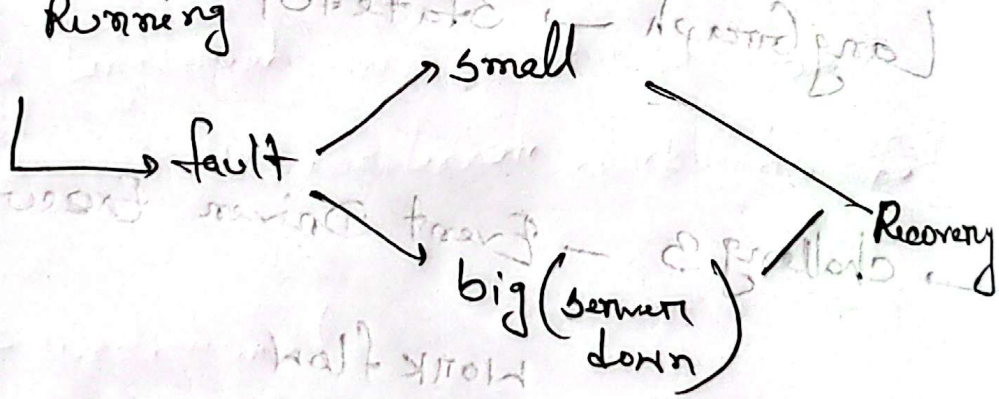
Prompt – LLM – Prompt – LLM – response

→ execute without stopping

* When a state fully execute they store the data at (check point) when external trigger is resume then the state resume in the next part.

→ challenge 4 – Fault Tolerance

Long Running

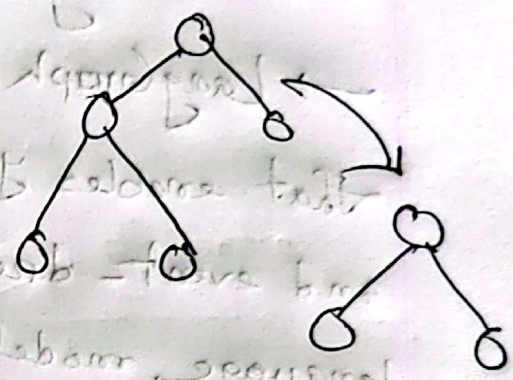


Long chain not exist in fault Tolerance.

→ challenge 5 – Human in the loop

Challenge 6 – Nested Workflows

↳ create subgraph



→ use multiagent

↳ reusability with the help of subgraph

Challenge 7 – Observability

↳ Observability refers to how easily you can monitor, debug and understand what your workflow is doing at runtime

Langchain → Langsmith ← llm application monitor

monitor

① What is LangGraph?

→ LangGraph is an orchestration framework that enables you to build stateful, multistep and event-driven workflows using large language models. it's ideal for designing both single and multi-agent Agentic AI.

* When to use what?

→ Use LangGraph when your use case involves complex, non-linear workflows that need:

conditional

Loops

Human-in-the-Loop steps

Multi-Agent coordination

Asynchronous or event-driven execution